

Requirements-Only PDF

Moodle Self-Hosted Implementation Requirements

CI Institute of Nursing | Implementation Requirements | Draft / Pending Final Internal Approval

Implementation/readiness requirements document.**No CDPH, CDSS, BRN, or government approval is claimed. Certificate generation remains gated until applicable approval metadata is issued and configured. Technical values marked for administrator verification must be confirmed before production install.**

Source	01_SELF_HOSTED_MOODLE_REQUIREMENTS.md
Output	01_SELF_HOSTED_MOODLE_REQUIREMENTS.pdf
Scope	Requirements-only; full Moodle implementation binder was not rebuilt.
Brand tokens	Maroon #8B1515 and gold #FFC107, based on the CI dynamic PDF studio reference.

1. Purpose and Scope

This requirements-only PDF converts the Moodle self-hosting requirements source file into a clean implementation readiness document for CI Institute of Nursing. It is limited to the Moodle self-host package and does not rebuild or alter the full implementation binder.

Status: Draft / Pending Approval. Generated Moodle MVP implementation artifact where source technical values were not available locally. Items marked "Technical admin final verification required" must be confirmed by the technical administrator before production install.

Approval boundary: This document does not claim CDPH, CDSS, BRN, or other government approval. Any certificate generation, approval metadata, or approval status remains Draft / Pending Approval or Pending Issuance until applicable approval data is issued and configured.

2. Requirement Summary Table

Requirement domain	Primary source controls	Readiness posture
Hosting / environment	Moodle version, web server, PHP/database, HTTPS/domain, SMTP, storage	Technical admin final verification required before production install.
Course architecture	Plugin policy, theme/branding, course-shell references	Use conservative CI Institute of Nursing branding; no approval-claim language.
Role / access	Authentication, roles, permissions, reviewer account controls	Least privilege; reviewer accounts read-only, scoped, time-limited.
Completion tracking	Cron and completion evidence retention	Cron must run reliably for completion processing and scheduled tasks.
Quiz / assessment	Evidence retention of grade reports and quiz attempts	Retain successful and failed attempt records as survey/audit evidence.
Certificate control	Certificate plugin policy and restricted administrative permissions	Certificate plugin remains disabled until approval metadata is issued and configured.
Reporting / audit export	Logging, retention/export, evidence retention	Logs and reports must be exportable for survey/audit review.
Data retention / backup / security	Backups, privacy/security hardening, retention/export, disaster recovery	Retention periods require SME/legal verification.
No-PHI / privacy	No-PHI controls and privacy/security hardening	Use fictional, de-identified scenarios only; quarantine/delete accidental PHI.
Accessibility / mobile	Accessibility and theme/branding controls	WCAG-oriented configuration with accessible theme and content practices.
Reviewer / survey access	Reviewer account controls and authentication	Reviewer access is temporary, read-only, scoped, and deactivated after survey window.
Go-live hold / release	Technical verification, named admin ownership, approval boundary	Hold release until technical and human verification items are closed.

3. Hosting and Environment Requirements

The hosting environment must be confirmed against the selected Moodle release at installation time. The requirements source deliberately avoids hard-coded unverified version claims.

Source requirement	Implementation requirement	Verification / control
Moodle version recommendation	- Default recommendation: Moodle 4.5 LTS for survey-stable self-hosting and plugin compatibility (long-term support release lifecycle reduces forced upgrades during a survey window). - Moodle 5.2 may be considered only if all required plugins, import tools (GIFT/CSV question import), themes, and the certificate plugin are verified compatible. - Version recommendation requires final technical verification before production install. Do not make unsupported version/security-support claims unless verified from current Moodle documentation. - Action: Technical admin final verification required - confirm the exact current LTS version and its supported PHP/database matrix from current official Moodle documentation at install time.	- Version recommendation requires final technical verification before production install. Do not make unsupported version/security-support claims unless verified from current Moodle documentation. - Action: Technical admin final verification required - confirm the exact current LTS version and its supported PHP/database matrix from current official Moodle documentation at install time.
Web server requirement	- A supported web server (e.g., Apache or Nginx) configured for the chosen Moodle version. - HTTPS enforced (see below). - Technical admin final verification required - confirm exact web server version against the Moodle release's requirements.	- Technical admin final verification required - confirm exact web server version against the Moodle release's requirements.
PHP / database requirement	- PHP version and required PHP extensions as specified for the chosen Moodle release. - A supported database engine (e.g., MariaDB/MySQL or PostgreSQL) at a version supported by the chosen Moodle release. - Do not hard-code specific version numbers here that are not verified. Technical admin final verification required - confirm PHP version, extensions, and database engine/version from the chosen release's official requirements page.	- Do not hard-code specific version numbers here that are not verified. Technical admin final verification required - confirm PHP version, extensions, and database engine/version from the chosen release's official requirements page.
HTTPS / domain requirement	- A dedicated domain or subdomain for the LMS. - Valid TLS certificate; force HTTPS site-wide; set wwwroot to the https URL. - Technical admin final verification required - domain ownership and certificate issuance.	- Technical admin final verification required - domain ownership and certificate issuance.
SMTP / email requirement	- Working outbound email (SMTP) for account creation, password reset, and notifications. - A monitored no-reply/support mailbox. - Technical admin final verification required - SMTP host/credentials.	- Technical admin final verification required - SMTP host/credentials.
Storage	- Adequate disk for moodledata, database, and backups, with growth headroom. - Technical admin final verification required - sizing based on expected enrollment and media.	- Technical admin final verification required - sizing based on expected enrollment and media.

4. Moodle Course Architecture Requirements

The requirements source limits architecture controls to verified Moodle configuration posture: minimal compatible plugins, maintained accessible theme, and conservative CI Institute of Nursing branding with no pre-approval claims.

Source requirement	Implementation requirement	Verification / control
Plugin policy	- Minimize plugins. Permitted/expected: a certificate plugin (e.g., a maintained Custom Certificate plugin) kept DISABLED until approval; optionally a configurable reports plugin. - Only install plugins verified compatible with the chosen Moodle version and from trusted sources. Technical admin final verification required for each plugin.	- Minimize plugins. Permitted/expected: a certificate plugin (e.g., a maintained Custom Certificate plugin) kept DISABLED until approval; optionally a configurable reports plugin. - Only install plugins verified compatible with the chosen Moodle version and from trusted sources. Technical admin final verification required for each plugin.
Theme / branding	- A maintained, accessible theme; CI Institute of Nursing branding applied conservatively. - No approval-claim language in branding before approval. See 17_ACCESSIBILITY_AND_MOBILE_REQUIREMENTS.md.	- No approval-claim language in branding before approval. See 17_ACCESSIBILITY_AND_MOBILE_REQUIREMENTS.md.

5. Role and Access Requirements

Source requirement	Implementation requirement	Verification / control
Authentication	- Manual accounts and/or email-based self-registration controlled by admin policy. - Strong password policy; reviewer accounts use temporary credentials with forced reset (see 04_ROLE_ACCESS_AND_WORKFLOW_MATRIX.md and 15_REVIEWER_ACCESS_AND_SURVEY_WORKFLOW.md). - Multi-factor authentication for privileged accounts where supported. Technical admin final verification required.	- Multi-factor authentication for privileged accounts where supported. Technical admin final verification required.
Roles	- Use the role set defined in 04_ROLE_ACCESS_AND_WORKFLOW_MATRIX.md and import_assets/ROLE_MATRIX_IMPORT_REFERENCE.csv, including a custom read-only External Survey Reviewer role.	- Configured and retained as implementation evidence before release.
Permissions	- Least-privilege configuration. Only Site Administrator / Manager may change site configuration, enable certificates, or perform documented grade overrides.	- Configured and retained as implementation evidence before release.

6. Completion-Tracking Requirements

Completion processing depends on reliable Moodle scheduled task execution and retention of completion evidence. Certificate issuance remains a post-approval control, not a pre-approval claim.

Source requirement	Implementation requirement	Verification / control
Cron requirement	- Moodle cron must run on a reliable schedule (e.g., every 1 minute) for completion processing, notifications, scheduled tasks, and backups. - Technical admin final verification required - cron configured and monitored.	- Technical admin final verification required - cron configured and monitored.

7. Quiz and Assessment Requirements

The source requirement explicitly requires retention of grade reports and quiz attempts, including failed attempts, as secure evidence. This PDF does not add unverified assessment configuration values beyond the source requirement.

Control	Requirement	Boundary
Assessment evidence	Retain grade reports and quiz attempts, including failed attempts, in the secure evidence archive.	Exact retention period requires SME/legal verification.

8. Certificate-Control Requirements

Certificate gating rule: Certificate generation must remain gated until applicable approval metadata is issued and configured. No government approval is claimed by this requirements document.

Control	Requirement	Release hold
Certificate plugin	A maintained certificate plugin may be expected, but it must remain disabled until approval.	Do not enable certificates or approval metadata until applicable approval data is issued and configured.
Administrative authority	Only Site Administrator / Manager may enable certificates or perform documented grade overrides.	Least-privilege permissions must be maintained.

9. Reporting and Audit-Export Requirements

Source requirement	Implementation requirement	Verification / control
Logging	- Standard Moodle logging enabled; logs retained per retention policy; log export available for survey/audit (see 13_REPORTING_AND_AUDIT_EXPORT_SPEC.md).	- Configured and retained as implementation evidence before release.
Evidence retention	- Retain completion reports, grade reports, quiz attempts (including failed attempts), logs, certificate issuance logs (post-approval), and course backups in a secure evidence archive for the required retention period (SME/legal verification required for exact periods).	- Retain completion reports, grade reports, quiz attempts (including failed attempts), logs, certificate issuance logs (post-approval), and course backups in a secure evidence archive for the required retention period (SME/legal verification required for exact periods).

10. Data Retention, Backup, and Security Requirements

Source requirement	Implementation requirement	Verification / control
Backups	- Automated scheduled backups of the database and moodledata, plus Moodle automated course backups (.mbz). - Offsite/secondary copy. See 14_DATA_RETENTION_BACKUP_AND_SECURITY_SPEC.md.	- Configured and retained as implementation evidence before release.
Privacy / security hardening	- Force HTTPS; secure cookies; disable unused features; restrict admin to dedicated accounts; keep Moodle and server patched. - Configure Moodle privacy/GDPR-style tools and a privacy policy page. Technical admin final verification required - jurisdiction-specific privacy requirements.	- Configure Moodle privacy/GDPR-style tools and a privacy policy page. Technical admin final verification required - jurisdiction-specific privacy requirements.
Retention / export	- Define retention for completion evidence and learner records aligned with each lane's recordkeeping requirements (see 14_DATA_RETENTION_BACKUP_AND_SECURITY_SPEC.md). Exact retention periods: SME/legal verification required.	- Define retention for completion evidence and learner records aligned with each lane's recordkeeping requirements (see 14_DATA_RETENTION_BACKUP_AND_SECURITY_SPEC.md). Exact retention periods: SME/legal verification required.
Disaster recovery	- Documented restore procedure; periodic restore test (see 08_SURVEY_READINESS_QA_TEST_PLAN.md, test BR-01). Technical admin final verification required - DR runbook.	- Documented restore procedure; periodic restore test (see 08_SURVEY_READINESS_QA_TEST_PLAN.md, test BR-01). Technical admin final verification required - DR runbook.

11. No-PHI and Privacy Requirements

Source requirement	Implementation requirement	Verification / control
No-PHI controls	- Learner warnings, activity prompt rules, and a quarantine/deletion process for any PHI accidentally entered. Fictional, de-identified scenarios only. See 16_NO_PHI_AND_PRIVACY_CONTROLS.md.	- Configured and retained as implementation evidence before release.

12. Accessibility and Mobile Requirements

Source requirement	Implementation requirement	Verification / control
Accessibility	- WCAG-oriented configuration (headings, alt text, contrast, captions/transcripts). See 17_ACCESSIBILITY_AND_MOBILE_REQUIREMENTS.md.	- Configured and retained as implementation evidence before release.

13. Reviewer and Survey Access Requirements

Source requirement	Implementation requirement	Verification / control
Reviewer account controls	- External Survey Reviewer accounts are read-only, time-limited, forced to reset password on first login, scoped to assigned categories/courses, and deactivated/expired after the survey window. See 15_REVIEWER_ACCESS_AND_SURVEY_WORKFLOW.md.	- Configured and retained as implementation evidence before release.

14. Go-Live Hold and Release Requirements

Go-live readiness remains held until technical administrator verification, human owner assignment, plugin compatibility review, privacy/security setup, and approval-boundary configuration are complete.

Source requirement	Implementation requirement	Verification / control
Admin ownership	- A named Site Administrator/Manager owns the environment, configuration, and approvals. (Named human owner assignment: final human verification required - do not fabricate a person.)	- A named Site Administrator/Manager owns the environment, configuration, and approvals. (Named human owner assignment: final human verification required - do not fabricate a person.)

15. Open Items / Human-Verification Items

Source area	Open verification item	Closeout rule
Moodle version recommendation	Action: Technical admin final verification required - confirm the exact current LTS version and its supported PHP/database matrix from current official Moodle documentation at install time.	Confirm before production install or go-live release; retain evidence of decision.
Web server requirement	Technical admin final verification required - confirm exact web server version against the Moodle release's requirements.	Confirm before production install or go-live release; retain evidence of decision.
PHP / database requirement	Do not hard-code specific version numbers here that are not verified. Technical admin final verification required - confirm PHP version, extensions, and database engine/version from the chosen release's official requirements page.	Confirm before production install or go-live release; retain evidence of decision.
HTTPS / domain requirement	Technical admin final verification required - domain ownership and certificate issuance.	Confirm before production install or go-live release; retain evidence of decision.
SMTP / email requirement	Technical admin final verification required - SMTP host/credentials.	Confirm before production install or go-live release; retain evidence of decision.
Cron requirement	Technical admin final verification required - cron configured and monitored.	Confirm before production install or go-live release; retain evidence of decision.
Storage	Technical admin final verification required - sizing based on expected enrollment and media.	Confirm before production install or go-live release; retain evidence of decision.
Authentication	Multi-factor authentication for privileged accounts where supported. Technical admin final verification required.	Confirm before production install or go-live release; retain evidence of decision.
Plugin policy	Only install plugins verified compatible with the chosen Moodle version and from trusted sources. Technical admin final verification required for each plugin.	Confirm before production install or go-live release; retain evidence of decision.
Privacy / security hardening	Configure Moodle privacy/GDPR-style tools and a privacy policy page. Technical admin final verification required - jurisdiction-specific privacy requirements.	Confirm before production install or go-live release; retain evidence of decision.

Source area	Open verification item	Closeout rule
Retention / export	Define retention for completion evidence and learner records aligned with each lane's recordkeeping requirements (see 14_DATA_RETENTION_BACKUP_AND_SECURITY_SPEC.md). Exact retention periods: SME/legal verification required.	Confirm before production install or go-live release; retain evidence of decision.
Disaster recovery	Documented restore procedure; periodic restore test (see 08_SURVEY_READINESS_QA_TEST_PLAN.md, test BR-01). Technical admin final verification required - DR runbook.	Confirm before production install or go-live release; retain evidence of decision.
Admin ownership	A named Site Administrator/Manager owns the environment, configuration, and approvals. (Named human owner assignment: final human verification required - do not fabricate a person.)	Confirm before production install or go-live release; retain evidence of decision.
Evidence retention	Retain completion reports, grade reports, quiz attempts (including failed attempts), logs, certificate issuance logs (post-approval), and course backups in a secure evidence archive for the required retention period (SME/legal verification required for exact periods).	Confirm before production install or go-live release; retain evidence of decision.

16. Final Requirement Validation

Validation control	Pass condition	Approval boundary
Source file	Requirements PDF is generated from 01_SELF_HOSTED_MOODLE_REQUIREMENTS.md only.	No full binder rebuild performed.
Branding	CI Institute of Nursing branding uses maroon #8B1515 and gold #FFC107 on clean letter-size pages.	No government approval claim appears.
Certificates	Certificate controls remain disabled/gated until applicable approval metadata is issued and configured.	Draft / Pending Approval or Pending Issuance only.
Access	Reviewer accounts are read-only, scoped, time-limited, and deactivated after survey window.	Least-privilege model retained.
Evidence	Logs, reports, grade reports, quiz attempts, completion reports, and certificate issuance logs post-approval are retained in secure evidence archive.	Exact retention periods require SME/legal verification.
Security/privacy	No-PHI controls, HTTPS, secure cookies, patching, privacy tools, backup, and DR requirements remain active go-live prerequisites.	Technical admin final verification required.

Final status: Draft / Pending Final Internal Approval. This requirements document supports implementation readiness planning and does not constitute regulatory approval.